Question 1:

A project has the following activities and dependencies with their optimistic (O), most likely (M), and pessimistic (P) times (in days):

Activity	Predecessor(s)	O	M	P
A	-	2	4	6
B	-	3	6	9
C	A	4	5	8
D	B	2	4	6
E	C,D	3	6	9

a) Calculate the expected time (TE) for each activity.

b) Draw the PERT chart.

c) Calculate ES, EF, LS, LF for each activity.

d) Find the critical path and project duration.

Question 2:

A software development project consists of the following activities with optimistic (O), most likely (M), and pessimistic (P) times (in weeks):

Activity	Predecessor(s)	O	M	P
A	-	1	2	3
B	A	2	4	6
C	A	1	3	5
D	B	3	5	7
E	C	2	4	6
F	D,E	1	2	3

a) Calculate the expected time (TE) for each activity.

b) Draw the PERT chart.

c) Calculate ES, EF, LS, LF for each activity.

d) Find the critical path and project duration.

Question 3:

A construction project has the following activities with estimated times:

Activity	Predecessor(s)	O	M	P
A	-	5	7	9
B	-	3	5	15
C	A	6	8	10
D	B	4	5	7
E	C, D	5	9	12

a) Calculate the expected time (TE) for each activity.

b) Draw the PERT chart.

c) Calculate ES, EF, LS, LF for each activity.

d) Find the critical path and project duration.

Question 4:

A product launch project has these activities and time estimates (in days):

Activity	Predecessor(s)	O	M	P
A	-	2	4	6
B	A	3	5	9
C	A	1	2	3
D	B	2	3	4
E	C	4	6	8
F	D,E	3	5	7

a) Calculate the expected time (TE) for each activity.

b) Draw the PERT chart.

c) Calculate ES, EF, LS, LF for each activity.

d) Find the critical path and project duration.